Neoantimon2-Converters — predictions with traceable origins

Nine published format adapters / eight method families • BA, EL and IM keep their scientific meaning

Retain the completed Neoantimon2 project — native prediction scores and grades are not required

1 CORE

Reconstruct

Variants + patient HLA
Explicit sample identity

Event → Origin → Peptide → Candidate

Many paths can share a peptide; all origin links are retained.

2 EXPORT

Export Bundle

neo_export_bundle()

Unique peptides / peptide–HLA pairs

Origin annotations + event associations

Keep project context.

3 RUN MODEL

Separate engines

BA · Binding affinity (e.g. MHCflurry)

EL · Presentation / eluted ligand

IM · Immunogenicity

User runs separately.

4 CONVERT

Registry adapters

All IDs below @1.0.0

BA: mhcflurry-ba

EL: bigmhc-el
mixmhcpred-el
mixmhc2pred-el
caphla-el

IM: bigmhc-im / prime-im
deepimmuno-cnn
immuneapp-im

Native output → standard TSV

5 INTERPRET

Import + explain

neo_analyze_external()
Strict peptide or peptide + HLA match

neo_external_results()
neo_external_explain()

Independent ranks + origin provenance

SCIENTIFIC AND EXECUTION BOUNDARIES

Adapters do not run engines or bundle binaries / weights. Review scripts and authorize execution.
BA, EL and IM scales differ. Raw IM scores are not EL percentiles and do not imply somatic A–E grades.
Native evidence, grades and ranks remain unchanged. Every matched candidate keeps all original Origin links.

Validation: converter + import tests used SCHEMA_FAITHFUL_SYNTHETIC_FIXTURE outputs.

Upstream predictor execution / model accuracy / clinical performance were not validated.